Lab 4: Sequence Diagrams

Project: eRegistrar
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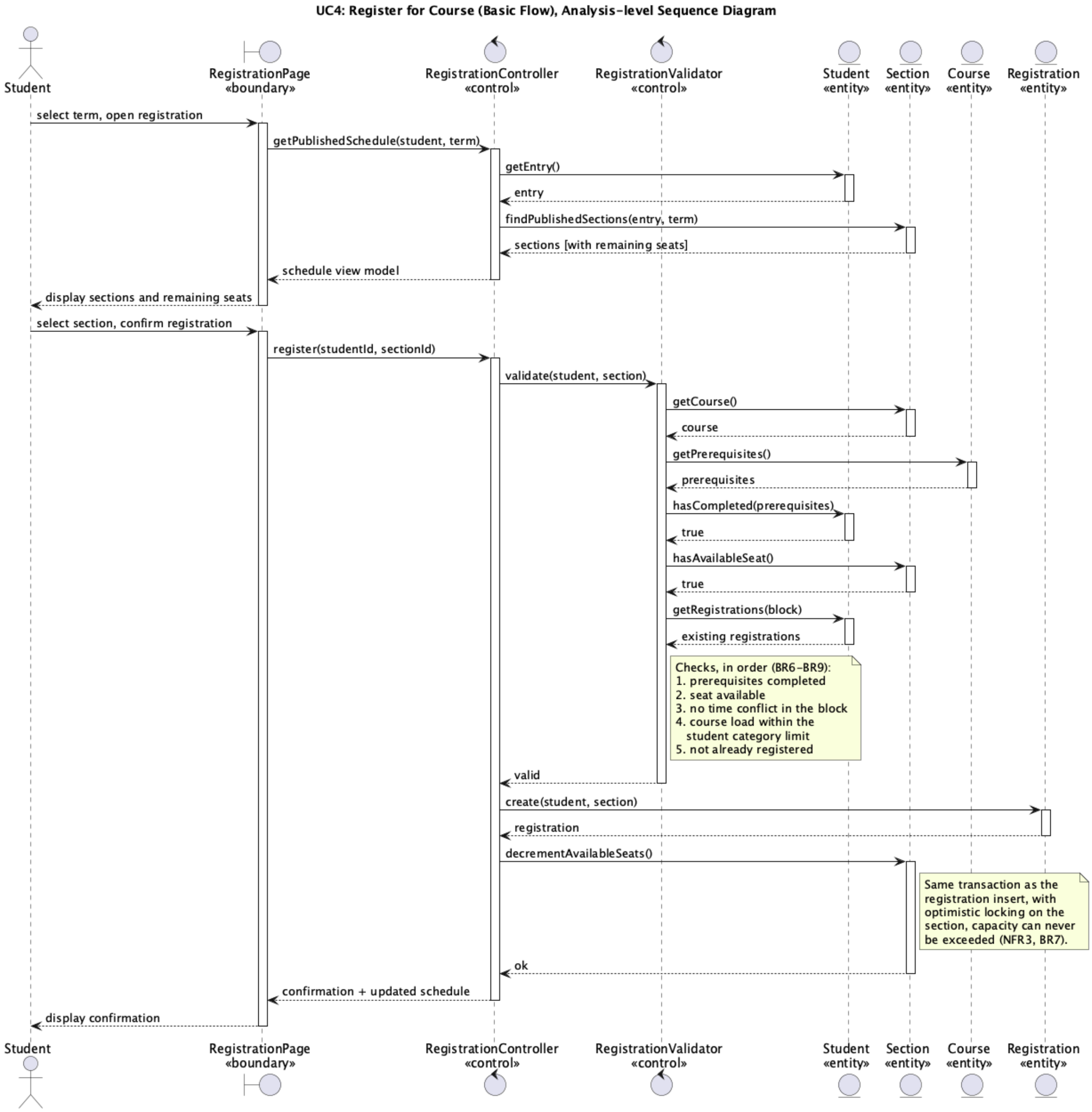
Sequence diagrams for the two major use cases of the SRS (Lab 2), drawn at the analysis level. Every lifeline is stereotyped as **«boundary»**, **«control»** or **«entity»**, and messages flow actor → boundary → control → entity, never back up.

Stereotype	Role
«boundary»	Mediates between an actor and the system; holds no business rules.
«control»	Coordinates the use case and applies the business rules.
«entity»	Long-lived stored information.

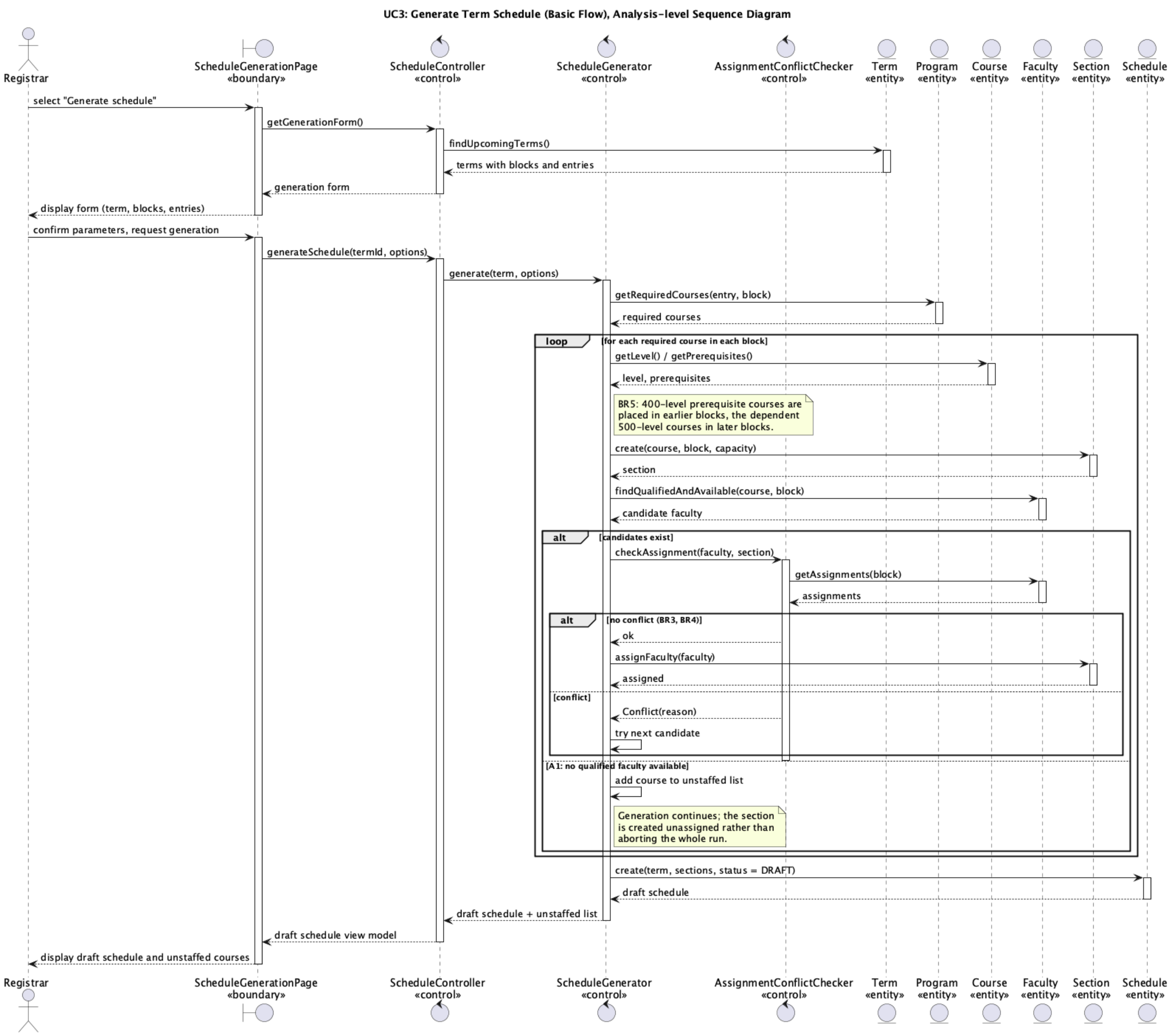
UC4: Register for Course



Class	Stereotype	Responsibility
`RegistrationPage`	boundary	Shows the schedule with remaining seats; collects the student's selection.
`RegistrationController`	control	Coordinates the use case; owns the transaction boundary.
`RegistrationValidator`	control	Applies BR6 to BR9: prerequisites, capacity, time conflict, course load.
`Student`, `Section`, `Course`, `Registration`	entity	The registration data.

The key detail is that the seat check and the seat decrement happen in the same transaction on `Section`, under optimistic locking. That's what makes NFR3 (capacity holds under concurrent registration) achievable.

UC3: Generate Term Schedule



Class	Stereotype	Responsibility
`ScheduleGenerationPage`	boundary	Generation form; displays the draft schedule and unstaffed courses.
`ScheduleController`	control	Coordinates the use case.
`ScheduleGenerator`	control	Determines required courses, creates sections, selects faculty (BR5).
`AssignmentConflictChecker`	control	Enforces BR3 (no double booking) and BR4 (only teachable courses).
`Term`, `Program`, `Course`, `Faculty`, `Section`, `Schedule`	entity	The scheduling data.

Alternate flow A1 shows up in the diagram: when no qualified, available faculty member exists, the section is created unassigned and the course is added to the unstaffed list instead of aborting the run.